

NAO ROS4HRI bridge workspace: modular ROS 2 graph

LLM backend(s)

chatbot model
planner model
Ollama / vLLM

NAO / simulator

nao_say_skill
nao_replay_motion
nao_look_at
NAOqi / sim

Human /
operator
speech,
rqt, traces

Dialogue and semantic routing

dialogue_manager
turn lifecycle
speech authority

chatbot_llm
route, response
planner handoff

Planning and goal handling

nao_orchestrator
planner gate
admission

planner_llm
plan generation
feedback response

Execution boundary

nao_orchestrator
validate plan
dispatch steps

Planner contract bus: /nao_orchestrator/planner_request → /planner/request → / intents → /planner/execution_feedback → /planner/dialogue_act

Grounding and symbolic evidence

nao_scene_grounding
detector facts
/scene/summary

kb_skills
KnowledgeCore
query/revise

grounded_context
entities, locations
bounded
relations

Evidence bus: /scene/summary, /kb/query, /kb/revise

Runtime skills and validation seams

/skill/scan
fresh evidence

/skill/
report_result
summary

/skill/move_to
bounded local

/skill/look_at
gaze

Validation, observability, and extension tracks

interaction_trace
viewer
JSON/HTML traces

robot-runtime
performance-review
main + fake policies

interaction_sim
perception +
KB fixtures

World Model Enricher
future temporal state
and uncertainty

- control / ROS contract
- grounding / evidence
- feedback / goal-state handling
- action / adapter call
- future extension